

para $x \neq 1$

① a) $\int dx = x + C$ b) $\int x^n = \frac{x^{n+1}}{n+1} + C$ c) $\int \frac{1}{x} dx = \int x^{-1} dx = \ln|x| + C$

$\rightarrow$ posso integrar cada termo.

d) $\int (3x^2 + 6x + 1) dx = x^3 + 3x^2 + x + C$ e) $\int \sqrt{x} dx = \int x^{\frac{1}{2}} dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2}{3} x^{\frac{3}{2}}$ f) $\int x^{-\frac{2}{5}} = \frac{x^{-\frac{2}{5}+1}}{-\frac{2}{5}+1} = \frac{x^{\frac{3}{5}}}{\frac{3}{5}}$

g) $\int (x^2 + 3x + 5 - 2x^{-1} + 4x^{-2}) dx = \frac{x^3}{3} + \frac{3x^2}{2} + 5x - 2\ln|x| - 4x^{-1} + C$ $\int \frac{2}{x} dx = 2\ln|x|$ $\int 4x^{-2} = \frac{4x^{-1}}{-1}$

h) $\int \frac{1}{2} \sqrt{x} \left(x + \frac{1}{x} \right) dx = \frac{1}{2} \int x^{\frac{1}{2}} \left(x + x^{-1} \right) dx = \frac{1}{2} \int x^{\frac{1}{2}} \cdot x + x^{\frac{1}{2}} \cdot x^{-1} dx = \frac{1}{2} \int x^{\frac{3}{2}} + x^{-\frac{1}{2}} dx = \left[\frac{x^{\frac{5}{2}}}{\frac{5}{2}} + x^{\frac{1}{2}} \cdot 2 \right] = \left(\frac{2}{5} \cdot 12 + 2\sqrt{2} \right) - \left(\frac{2}{5} + 2 \right)$

i) $\int \frac{5x^2}{x^3} + \frac{7x^{\frac{1}{3}}}{x^{\frac{1}{3}}} + 2x^{-\frac{4}{3}} dx = \int 5x^{-\frac{1}{3}} + x^{-1} + 2x^{-\frac{4}{3}} dx = \frac{5x^{\frac{2}{3}}}{\frac{2}{3}} + \ln|x| + 2x^{-\frac{1}{3}} = 3x^{\frac{2}{3}} + \ln|x| - 6x^{-\frac{1}{3}}$

j) $\int \frac{1}{1+x^2} dx = \arctg x + C$

② $f(x) = \frac{-1}{\sqrt{x}} + 3x^2 - 5 = -x^{-\frac{1}{2}} + 3x^2 - 5$ $\int -x^{-\frac{1}{2}} + 3x^2 - 5 = -\int x^{-\frac{1}{2}} dx + 3\int x^2 dx - \int 5 = \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + 3 \cdot \frac{x^3}{3} - 5x = \frac{2}{\sqrt{x}} + x^3 - 5x = \boxed{\frac{2}{\sqrt{x}} + x^3 - 5x + C}$

para satisfazer $F(1) = 3 \rightarrow 2 + 1 - 5 + C = 3 \rightarrow C = 5$ $F(x) = \frac{2}{\sqrt{x}} + x^3 - 5x + 5$

③ a) $\int \frac{1+x^2}{x^2} dx = \int \frac{1}{x^2} dx + \int 1 dx = -\frac{1}{x} + x + C$ $\int x^{-2} dx = \frac{x^{-1}}{-1}$

b) $\frac{4x^5 + 0x^4 - 4x^3 - 2x^2 + 0x + 1}{-4x^5 + 0x^4 - 4x^3} \Big| \frac{x^2 + 0x + 1}{4x^3 - 8x - 2} = 4x^3 - 8x - 2 + \frac{8x + 2}{x^2 + 1}$

c) $\frac{2x^5 + 0x^4 + 14x^3 - 0x^2 - 36x + 6}{-2x^5 - 0x^4 - 18x^3} \Big| \frac{x^2 + 0x + 1}{2x^3 - 4x} = \frac{-4x^3 - 0x^2 - 36x}{14x^3 + 0x^2 + 36x} = \frac{6}{6}$

$\frac{-2x^3 + 8x + 1}{12x^3 + 0x + 1} = 8x + 3$

$\int 4x^3 - 8x - 2 + \frac{8x + 2}{x^2 + 1} = \int 4x^3 - 8x - 2 + \frac{8x + 2}{x^2 + 1} =$

$= 4 \cdot \frac{x^4}{4} - 8 \cdot \frac{x^2}{2} - 2x + \int \frac{8x}{x^2 + 1} + \int \frac{2}{x^2 + 1} = \boxed{x^4 - 4x^2 - 2x + 4\ln|x| + 2\arctg(x)}$

① $\int \frac{8x}{x^2 + 1} = \int \frac{8x}{u} dx = \int \frac{4 du}{u} = 4\ln|u|$

$u = x^2 + 1$

$\frac{du}{dx} = 2x \Rightarrow du = 2x dx$ $dx \cdot 8x = 4 du$

② $\int \frac{3}{x^2 + 1} = 3 \cdot \int \frac{1}{x^2 + 1} = 3\arctg(x)$

d) $3 \int \frac{\sec^2(x) \cdot \cos^2(x)}{\sec^2(x)} + \sec^2(x) \cdot \cos^2(x) dx =$

$= 3 \int 1 + \frac{1}{\sec^2(x)} = 3x + \int \frac{1}{\sec^2(x)} = \boxed{-3x \cdot \cos^2(x) + C}$

$$\begin{cases} 2x+2, & \text{se } x \geq -1 \\ -(2x+2), & \text{se } x < -1 \end{cases}$$

g) $\int_{-2}^1 -2x-2 = -x^2-2x = -(x^2+2x) \quad [-x^2+2x]_{-2}^1 = -[1-4+4] = +1$

$\int_{-2}^1 |2x+2| \quad \int_{-1}^1 2x+2 = x^2+2x \quad [x^2+2x]_{-1}^1 = (1-2+4) = 9$

$$\int_{-2}^1 |2x+2| = 10$$

h) $\int_{\frac{\pi}{2}}^{\pi} |\sin x| dx = \int_{\frac{\pi}{2}}^{\pi} \sin x dx = [-\cos x]_{\frac{\pi}{2}}^{\pi} = (-\cos \pi) - (-\cos \frac{\pi}{2}) = 1$



i) $\int \sin x + 3 \cos x dx = \int \sin x dx + 3 \int \cos x dx = -\cos x + 3 \sin x + C$

j) $\int \frac{x^2+1}{x} dx = \int \frac{x^2}{x} dx + \int \frac{1}{x} dx = \frac{x^2}{2} + \frac{x}{x}$

k) $\int \frac{1}{\sin^2 x} dx = \int \frac{1}{\sin^2 x} dx = -\cot x + C$

5) a) $\int_0^{\frac{\pi}{2}} \cos x dx = [\sin x]_0^{\frac{\pi}{2}} = \sin \frac{\pi}{2} - \sin 0 = 1$

b) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x + \cos x dx = [-\cos x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + [\sin x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = -\cos \frac{\pi}{2} - \cos(-\frac{\pi}{2}) + \sin \frac{\pi}{2} - \sin(-\frac{\pi}{2}) = 2$

c) $\int_0^1 x^2 dx = \frac{1}{3} [x^3]_0^1 = \frac{1}{3} (1-0) = \frac{1}{3}$

d) $\int_0^{2\pi} \sin x dx = [-\cos x]_0^{2\pi} = -\cos 2\pi - (-\cos 0) = 0$

6) Teorema fundamental da calculo: $F(x) = \int_a^x f(t) dt \implies \frac{dF(x)}{dx} = f(x)$

a) $\int_1^x (5x-2) dx \implies \frac{dF(x)}{dx} = 5x-2$

b) $F(x) = \int_5^x \sqrt{x} dx \implies \frac{dF(x)}{dx} = \sqrt{x}$

MÉTODO DA SUBSTITUIÇÃO:

5) a) $\int 3x^2 \sqrt{x^3+1} dx = \int \sqrt{u} du = \frac{2}{3} u^{\frac{3}{2}} = \frac{2}{3} (x^3+1)^{\frac{3}{2}} + C$

$u = x^3+1 \quad \frac{du}{dx} = 3x^2 \quad du = 3x^2 dx$

b) $\int (x+1)^{1/3} dx = \frac{u^{4/3}}{4/3} = \frac{3}{4} (x+1)^{4/3} + C$

$u = x+1 \quad \frac{du}{dx} = 1 \quad du = dx$

c) $\int e^{3x} dx = \int e^u \frac{du}{3} = \frac{1}{3} \int e^u du = \frac{1}{3} e^u = \frac{1}{3} e^{3x} + C$

$u = 3x \quad du = 3 dx \quad \frac{du}{dx} = 3$

d) $\int \sin 7x dx = \int \sin u \frac{du}{7} = \frac{1}{7} \int \sin u du = \frac{-\cos u}{7} = \frac{-\cos 7x}{7} + C$

$u = 7x \quad \frac{du}{dx} = 7 \quad dx = \frac{du}{7}$

e) $\int x \cdot e^{x^2} dx = \int e^u du = e^u + C = e^{x^2} + C$

$u = x^2 \quad \frac{du}{dx} = 2x \quad du = 2x dx$

f) $\int \sec(2x) \cdot \tan(2x) dx = \int \sec(u) \cdot \tan(u) du = \frac{\sec(u)}{2} = \frac{\sec 2x}{2} + C$

$u = 2x \quad \frac{du}{dx} = 2 \quad du = 2 dx$

$\frac{d}{dx} \sec(u) = \sec(u) \cdot \tan(u)$

g) $\int e^{\cos x} \cdot \sin x dx = -\int e^u du = -e^u = -e^{\cos x}$

$u = \cos x \quad \frac{du}{dx} = -\sin x \quad du = -\sin x dx$

h) $\int \frac{1}{x+1} dx = \int \frac{1}{u} du = \ln|u| = \ln|x+1|$

$u = x+1 \quad \frac{du}{dx} = 1 \quad du = dx$

i) $\int \frac{3}{(1-x)^5} dx = \frac{3}{4} \int \frac{1}{u^5} du = \frac{3}{4} \int u^{-5} du = \frac{3}{4} \frac{u^{-4}}{-4} = \frac{-3}{16} (1-x)^{-4} = \frac{-3}{16(1-x)^4} + C$

$u = 1-x \quad \frac{du}{dx} = -1 \quad dx = -du$

$$g) \int \frac{x}{x^2+1} dx = \int \frac{x}{u} \cdot \frac{du}{2x} = \frac{1}{2} \int \frac{1}{u} du = \frac{\ln|u|}{2} = \frac{\ln|x^2+1|}{2} + C$$

$$u = x^2 + 1 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2x}$$

$$h) \int \sin(5x + \frac{\pi}{2}) dx = \frac{1}{5} \int \sin u du = -\frac{\cos(5x + \frac{\pi}{2})}{5} + C$$

$$u = 5x + \frac{\pi}{2} \quad \frac{du}{dx} = 5 \quad dx = \frac{du}{5}$$

$$l) \int_4^6 \sqrt{5x-1} dx = \frac{1}{5} \int_4^6 \sqrt{u} du = \frac{1}{5} \int_{19}^{29} u^{\frac{1}{2}} du = \frac{1}{5} \left[\frac{2}{3} u^{\frac{3}{2}} + C \right]_{19}^{29} = \frac{2}{75} [29\sqrt{29} - 19\sqrt{19}] + C$$

$$u = 5x - 1 \quad \frac{du}{dx} = 5 \quad dx = \frac{du}{5} \quad \text{quando: } x=4 \quad u=19 \quad x=6 \quad u=29$$

$$m) \int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = 2 \int \cos u du = 2 \sin \sqrt{x} + C$$

$$u = \sqrt{x} \quad x = u^2 \quad dx = 2u du$$

$$n) \int e^x \cdot \sin e^x dx = \int e^x \sin u \cdot \frac{du}{e^x} = \int \sin u du = -\cos u + C = -\cos e^x + C$$

$$e^x = u \quad \frac{du}{dx} = e^x \quad dx = \frac{du}{e^x}$$

$$o) \int \frac{\cos(\frac{\pi}{x})}{x^2} dx = -\frac{1}{\pi} \int \cos(u) \frac{du}{x^2} = -\frac{1}{\pi} \sin(\frac{\pi}{x}) + C$$

$$u = \frac{\pi}{x} = \pi x^{-1} \quad \frac{du}{dx} = -\pi x^{-2} \quad dx = \frac{du}{-\pi}$$

*troço da derivada

$$p) \int \frac{x}{1+x^4} dx = \int \frac{x}{1+u^2} \cdot \frac{du}{2x} = \frac{1}{2} \int \frac{1}{1+u^2} du = \frac{1}{2} \arctg u + C = \frac{1}{2} \arctg x^2 + C$$

$$\text{lembrando que } \int \frac{1}{1+x^2} = \arctg(x) \quad u = x^2 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2x}$$

$$r) \int \cos x \cdot \sin^6 x dx = \int u^6 du = \frac{\sin^7 x}{7} + C$$

$$u = \sin x \quad \frac{du}{dx} = \cos x \quad dx = \frac{du}{\cos x}$$

$$s) \int \frac{\ln|x|}{x} dx = \int \frac{u}{x} \frac{du}{x} = \frac{u^2}{2} = \frac{(\ln|x|)^2}{2} + C$$

$$u = \ln|x| \quad \frac{du}{dx} = \frac{1}{x} \quad dx = du \cdot x$$

$$t) \int 2x \sqrt{x^2+4} dx = \int 2x \sqrt{u} \frac{du}{2x} = \int u^{\frac{1}{2}} du = \frac{2}{3} u^{\frac{3}{2}} = \frac{2}{3} (x^2+4)^{\frac{3}{2}} + C$$

$$u = x^2 + 4 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2x}$$

$$u = x^2 + 4 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2x}$$

$$v) \int \frac{\tan x}{\cos x} dx = \int \frac{\sin x}{\cos^2 x} dx = -\int \frac{\sin x}{u} \frac{du}{-\sin x} = -\int \frac{1}{u} du = -\ln|u| = -\ln|\cos x| + C$$

$$w) \int \frac{\tan x}{\cos x} dx = \int \frac{\sin x}{\cos^2 x} dx = -\int \frac{\sin x}{u} \frac{du}{-\sin x} = -\int \frac{1}{u} du = -\ln|u| = -\ln|\cos x| + C$$

$$u) \int_0^3 x \sqrt{1+x} dx = \int_0^3 x u du = \int_1^4 (u-1) \sqrt{u} du = \int_1^4 (u-1) u^{\frac{1}{2}} du =$$

$$= \int_1^4 (\frac{3}{2} u^{\frac{3}{2}} - \frac{1}{2} u^{\frac{1}{2}}) du = \left[\frac{3}{2} \cdot \frac{2}{5} u^{\frac{5}{2}} - \frac{1}{2} \cdot \frac{2}{3} u^{\frac{3}{2}} \right]_1^4 = \frac{6}{5} u^{\frac{5}{2}} - \frac{1}{3} u^{\frac{3}{2}} \Big|_1^4$$

*Integro cada termo

$$\begin{aligned} u &= 1+x & \frac{du}{dx} &= 1 & du &= dx \\ x &= u-1 & & & & \\ \text{quando } x=0 &\Rightarrow u=1 \\ x=3 &\Rightarrow u=4 \end{aligned}$$

$$\int_0^3 x \sqrt{1+x} dx = \frac{116}{15}$$

$$u) \int_0^4 \sqrt{3x+4} dx = \frac{1}{3} \int_4^{16} \sqrt{u} du = \frac{1}{3} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_4^{16} = \frac{2}{9} \left[\frac{16^{\frac{3}{2}}}{2} - \frac{4^{\frac{3}{2}}}{2} \right] = \frac{256}{9} - \frac{16}{9} = \frac{240}{9} = \frac{80}{3}$$

$$u = 3x + 4 \quad \frac{du}{dx} = 3 \quad dx = \frac{du}{3}$$

$$\text{quando } x=0 \rightarrow u=4 \quad x=4 \rightarrow u=16$$

$$x) \int 2x^2 (4-5x^3)^5 dx = \int 2x^2 \cdot u^5 \cdot \frac{du}{-15x^2} = -\frac{2}{15} \int u^5 du = -\frac{2}{15} \frac{u^6}{6} = -\frac{u^6}{45} = -\frac{(4-5x^3)^6}{45} + C$$

$$u = 4-5x^3 \quad \frac{du}{dx} = -15x^2 \quad dx = \frac{du}{-15x^2}$$

$$y) \int x \cdot \tan(x^2) dx = \int x \cdot \tan(u) \frac{du}{2x} = -\frac{\cos x^2}{2}$$

$$u = x^2 \quad dx = \frac{du}{2x}$$

$$z) \int \sin x \cdot \sqrt{\cos x} dx = -\int u^{\frac{1}{2}} du = -\frac{u^{\frac{3}{2}}}{\frac{3}{2}} = -\frac{\cos^{\frac{3}{2}}(x) \cdot 2}{3}$$

$$u = \cos x \quad \frac{du}{dx} = -\sin x \quad dx = \frac{du}{-\sin x}$$

$$6) \int \sec x dx \cdot \frac{(\sec x + \tan x)}{\sec x + \tan x} = \int \frac{\sec^2 x + \sec x \tan x}{u} dx = \int \frac{\sec^2 x + \sec x \tan x}{u} \cdot \frac{du}{\sec(x) \tan(x) + \sec^2(x)} = \int \frac{1}{u} du = \ln|u| = *$$

$$u = \sec x + \tan x \quad \frac{du}{\sec(x) \tan(x) + \sec^2(x)} = dx$$

$$* \int \sec x dx = \ln|\sec(x) + \tan(x)| + C$$

INTEGRAL POR PARTES: $\int u dv = u \cdot v - \int v \cdot du$

$$a) \int x \cdot \cos 5x dx \quad \begin{matrix} \text{DERIVAR} & \text{L'INTEGRAR} \end{matrix} = x \cdot \frac{1}{5} \sin 5x - \int \frac{1}{5} \sin 5x dx = \frac{x}{5} \sin 5x - \frac{1}{5} \int \sin 5x dx = *$$

DERIVAR

$$* = \frac{x}{5} \sin 5x - \frac{1}{5} \left(-\frac{1}{5} \cos 5x \right) = \frac{x}{5} \sin 5x + \frac{\cos 5x}{25}$$

$$\begin{matrix} u = x \rightarrow du = dx \\ v = \cos 5x \rightarrow \frac{1}{5} \sin 5x \end{matrix}$$

$$b) \int x \cdot e^x dx = x \cdot e^x - \int e^x dx = x \cdot e^x - e^x = e^x (x-1) + C$$

$\begin{matrix} \text{L'INTEGRAR} \\ \text{DERIVAR} \end{matrix}$

$$\begin{matrix} u = x & dv = e^x dx \rightarrow v = e^x \\ \frac{du}{dx} = 1 \rightarrow du = dx & * \text{Integrando} \end{matrix}$$

$$d) \int \cos^4(x) dx = \int \cos^2(x) \cdot \cos^2(x) dx = \int \frac{(\cos(2x)+1)}{2} \cdot \frac{(\cos(2x)+1)}{2} dx = \int \frac{\cos^2(2x) + 2\cos(2x) + 1}{4} dx = *$$

$$* \int \frac{1}{4} dx + \int \frac{2\cos(2x)}{4} + \int \frac{\cos^2(2x)}{4} = \frac{1}{4} + \int \frac{\cos(2x)}{2} + \frac{1}{4} + \int \frac{\cos(4x)}{8} = \frac{3}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$$

$$\cos(2x) = \cos^2 x - \sin^2 x = \cos^2 x - 1 + \cos^2 x = 2\cos^2 x - 1$$

$$\cos(2x) = 2\cos^2(x) - 1 \rightarrow \cos^2(x) = \frac{\cos(2x) + 1}{2}$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos(4x) = \cos^2(2x) - \sin^2(2x) = 2\cos^2(2x) - 1$$

$$\rightarrow \cos^2(2x) = \frac{\cos(4x) + 1}{2}$$

$$\sin^2(2x) = 1 - \cos^2(2x)$$

$$\int \frac{\cos(2x)}{2} dx = \int \frac{\cos u}{2} \cdot \frac{du}{2} = \frac{1}{4} \sin(2x)$$

$$u = 2x \quad \frac{du}{dx} = 2 \quad dx = \frac{du}{2}$$

$$\int \frac{\cos(4x)}{8} dx = \int \frac{\cos(4x)}{8} \cdot \frac{du}{4} = \frac{1}{32} \sin(4x)$$

$$u = 4x \quad \frac{du}{dx} = 4 \quad dx = \frac{du}{4}$$

$$\int \cos^4(x) dx = \frac{3}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$$

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

$$e) \int \sin^3 x \, dx = \int \sin^2(x) \cdot \sin(x) \, dx = \int (1 - \cos^2(x)) \cdot \sin(x) \, dx = \int (1 - u^2) \cdot \sin(x) \, dx = \int (1 - u^2) \cdot du = \textcircled{I}$$

$$\sin^2 = 1 - \cos^2 x$$

$$\textcircled{I} - \int (1 - u^2) du = - \left(\int 1 du - \int u^2 du \right) = - \left(u - \frac{u^3}{3} \right) = - \left(u - \frac{u^3}{3} \right) = \frac{\cos^3(x)}{3} - \cos(x) + C$$

$$u = \cos x$$

$$\frac{du}{dx} = -\sin x \quad \therefore -du = \sin x \cdot dx$$

$$g) \int \ln(x) \, dx = x \cdot \ln(x) - \int x \cdot \frac{du}{x} = x \cdot \ln(x) - \int 1 du = x \cdot \ln(x) - x + C$$

$$u = \ln(x) \rightarrow \frac{du}{dx} = \frac{1}{x} \quad \therefore dx = \frac{du}{x}$$

$$dv = 1 \rightarrow \int dv = \int 1 \quad \therefore v = x$$

$$g) \int x \ln(x) \, dx = \ln(x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{dx}{x} = \ln(x) \cdot \frac{x^2}{2} - \int \frac{x}{2} dx = \ln(x) \cdot \frac{x^2}{2} - \frac{1}{2} \cdot \frac{x^2}{2} = \frac{x^2}{2} \left(\ln(x) - \frac{1}{2} \right) + C$$

$$u = \ln(x) \rightarrow \frac{du}{dx} = \frac{1}{x} \quad du = \frac{dx}{x} \quad dv = x \xrightarrow{\text{INTEGRAL}} v = \frac{x^2}{2}$$

$$h) \int \sqrt{x} \ln(x) \, dx = \ln(x) \cdot \frac{x^{\frac{3}{2}}}{\frac{3}{2}} - \int \frac{x^{\frac{3}{2}}}{\frac{3}{2}} \cdot \frac{dx}{x} = \ln(x) \cdot \frac{2}{3} x^{\frac{3}{2}} - \frac{2}{3} \int \frac{1}{x^{\frac{1}{2}}} dx = \ln(x) \cdot \frac{2}{3} x^{\frac{3}{2}} - \frac{2}{3} \cdot \frac{x^{\frac{1}{2}}}{\frac{1}{2}} = \ln(x) \cdot \frac{2}{3} x^{\frac{3}{2}} - \frac{4}{9} x^{\frac{3}{2}}$$

$$u = \ln(x) \rightarrow \frac{du}{dx} = \frac{1}{x}$$

$$dv = \sqrt{x} \xrightarrow{\text{INTEGRAL}} v = \frac{x^{\frac{3}{2}}}{\frac{3}{2}}$$

$$i) \int \cos \sqrt{x} \, dx = \int \cos \sqrt{u} \cdot 2u \, du = 2 \int \cos u \cdot u \, du = 2 \cdot (\sqrt{x} \cdot \sin \sqrt{x} + \cos \sqrt{x}) + C = 2\sqrt{x} \cdot \sin \sqrt{x} + 2 \cos \sqrt{x} + C$$

$$u = \sqrt{x} \rightarrow u^2 = x$$

$$\therefore \frac{dx}{du} = 2u \rightarrow dx = 2u \, du$$

$$\textcircled{I} \int \cos u \cdot 2u \, du = 2 \cdot \sin u - \int \sin(u) \, du = u \cdot \sin(u) + \cos(u) + C = \sqrt{x} \cdot \sin \sqrt{x} + \cos \sqrt{x} + C$$

$$x = u \quad \frac{dx}{du} = 1 \quad \therefore dx = du$$

$$dv = \cos u \rightarrow v = \sin u$$

$$\int \sin(u) \, du = -\cos(u)$$

$$\int \cos \sqrt{x} \, dx = 2\sqrt{x} \cdot \sin \sqrt{x} + 2 \cos \sqrt{x} + C$$

$$k) \int x \ln(3x) \, dx = \ln(3x) \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{dx}{x} = \frac{x^2}{2} \ln(3x) - \frac{1}{2} \int x \, dx = \frac{x^2}{2} \ln(3x) - \frac{1}{2} \cdot \frac{x^2}{2} = \frac{x^2}{2} \left(\ln(3x) - \frac{1}{2} \right) + C$$

$$u = \ln(3x) \quad \frac{du}{dx} = \frac{1}{3x} = \frac{1}{x}$$

$$dv = x \xrightarrow{\text{INTEGRAL}} v = \frac{x^2}{2}$$

$$d) \int x \cdot \cos(2x) dx = \frac{x \cdot \sin(2x)}{2} - \int \frac{\sin 2x}{2} dx = \frac{x \cdot \sin(2x)}{2} - \frac{1}{2} \int \sin 2x dx = \frac{x \cdot \sin(2x)}{2} - \frac{1}{2} \left(-\frac{1}{2} \cos(2x) \right) = \frac{x \cdot \sin(2x)}{2} + \frac{1}{4} \cos(2x) + C$$

$$u = x \rightarrow \frac{du}{dx} = 1 \therefore du = dx$$

$$dv = \cos(2x) \xrightarrow{\text{INTEGRAR}} v = \int \cos 2x dx = \int \cos t \frac{dt}{2} = \frac{1}{2} \sin t = \frac{\sin 2x}{2}$$

$$t = 2x \quad \frac{dt}{dx} = 2 \quad v = \frac{\sin 2x}{2}$$

$$\int x \cdot \cos(2x) dx = \frac{x \cdot \sin(2x)}{2} + \frac{1}{4} \cos(2x) + C$$

$$e) \int \sin 2x dx = \int \sin t \frac{dt}{2} = \frac{1}{2} - \cos t$$

$$t = 2x \quad \frac{dt}{dx} = 2 \quad dx = \frac{dt}{2}$$

Integral por frações parciais:

$$f) \int \frac{2x-1}{x^2-1} dx = \int \frac{2x-1}{(x-1)(x+1)} dx = \int \frac{\frac{2}{x-1}}{(x-1)} dx + \int \frac{\frac{-3}{x+1}}{(x+1)} dx = \frac{2}{2} \ln|x-1| + \frac{-3}{2} \ln|x+1| + C$$

$$\frac{2x-1}{(x-1)(x+1)} = \frac{A}{(x-1)} + \frac{B}{(x+1)} = \frac{A(x+1) + B(x-1)}{(x-1)(x+1)} = \frac{Ax + A + Bx - B}{(x-1)(x+1)} = \frac{x(A+B) - B + A}{(x-1)(x+1)}$$

$$\frac{2x-1}{(x-1)(x+1)} = \frac{x(A+B) - B + A}{(x-1)(x+1)}$$

$$\begin{cases} A+B=2 & A=\frac{2}{2} \\ A-B=-1 & B=\frac{3}{2} \end{cases} \quad \frac{A}{(x-1)} + \frac{B}{(x+1)} = \frac{\frac{2}{2}}{(x-1)} + \frac{\frac{3}{2}}{(x+1)}$$

$$g) \int \frac{3x-1}{(x-2)(x+5)} dx = \int \frac{\frac{5}{x-2}}{(x-2)} dx + \int \frac{\frac{16}{x+5}}{(x+5)} dx = \frac{5}{2} \ln|x-2| + \frac{16}{7} \ln|x+5| + C$$

$$\frac{3x-1}{(x-2)(x+5)} = \frac{A}{(x-2)} + \frac{B}{(x+5)} = \frac{A(x+5) + B(x-2)}{(x-2)(x+5)} \rightarrow 3x-1 = Ax+5A+Bx-2B \quad \begin{cases} A+B=3 & A=\frac{5}{7} \\ 5A-2B=-1 & B=\frac{16}{7} \end{cases}$$

$$h) \int \frac{2x+4}{x^3-2x^2} dx = \int \frac{2x+4}{x^2(x-2)} dx = \int \frac{A}{(x-2)} dx + \int \frac{Bx+C}{x^2} dx = 2 \ln|x-2| + \int \frac{-2x-2}{x^2} dx = 2 \ln|x-2| - 2 \ln|x| + \frac{2}{x} + C$$

$$\frac{2x+4}{x^2(x-2)} = \frac{A}{(x-2)} + \frac{Bx+C}{x^2} = \frac{Ax^2 + Bx + C}{x^2(x-2)} = \frac{Ax^2 + Bx^2 - 2Bx + Cx - 2C}{x^2(x-2)} = \frac{x^2(A+B) - x(2B-C) - 2C}{x^2(x-2)}$$

$$0x^2 + 2x + 4 = x^2(A+B) - x(2B-C) - 2C$$

$$\begin{cases} A+B=0 \\ 2B-C=2 \\ -2C=4 \end{cases} \quad \begin{cases} A=2 \\ B=-2 \\ C=-2 \end{cases}$$

$$i) \int \frac{-2x-2}{x^2} dx = \int \frac{-2}{x} dx + \int \frac{-2}{x^2} dx = -2 \int \frac{1}{x} dx + 2 \int x^{-2} dx = -2 \left(\ln|x| - \frac{1}{x} \right) = -2 \ln|x| + \frac{2}{x}$$

①

$$d \int \frac{5}{x \cdot (x^2-9)} dx = \int \frac{5}{x \cdot (x-3) \cdot (x+3)} = \int \frac{-\frac{5}{9}}{x} dx + \int \frac{\frac{5}{18}}{(x-3)} dx + \int \frac{\frac{5}{18}}{(x+3)} dx = -\frac{5}{9} \ln|x| + \frac{5}{18} \ln|x-3| + \frac{5}{18} \ln|x+3|$$

$$\textcircled{I} \frac{0x^2+0x+5}{x \cdot (x-3) \cdot (x+3)} = \frac{A}{x} + \frac{B}{(x-3)} + \frac{C}{(x+3)} = \frac{(x^2-9)A+B(x^2+3x)C(x^2-3x)}{x \cdot (x-3) \cdot (x+3)} = \frac{Ax^2-9A+Bx^2+3Bx+Cx^2-3Cx}{x \cdot (x-3) \cdot (x+3)}$$

$$0x^2+0x+5 = x^2(A+B+C) + x(3B-3C) - 9A$$

$$\begin{cases} A+B+C=0 \rightarrow 2B=\frac{5}{9} \rightarrow B=\frac{5}{18} \\ 3B-3C=0 \rightarrow B=C \quad C=\frac{5}{18} \\ -9A=5 \quad A=-\frac{5}{9} \end{cases}$$

$$e) \int \frac{1}{x^3+x} dx = \int \frac{1}{x \cdot (x^2+1)} dx = \int \frac{1}{x} dx + \int \frac{-x}{(x^2+1)} dx = \ln|x| - \frac{1}{2} \ln|x^2+1| + C$$

$$\textcircled{I} \frac{1}{x \cdot (x^2+1)} = \frac{A}{x} + \frac{Bx+C}{(x^2+1)} = \frac{A \cdot (x^2+1) + x(Bx+C)}{x \cdot (x^2+1)} = \frac{Ax^2+A+x^2B+xC}{x \cdot (x^2+1)} = \frac{x^2 \cdot (A+B) + xC + A}{x \cdot (x^2+1)}$$

$$\frac{0x^2+0x+1}{x \cdot (x^2+1)} = \frac{x^2 \cdot (A+B) + xC + A}{x \cdot (x^2+1)} \quad \begin{cases} A+B=0 \\ C=0 \\ A=1 \end{cases} \quad B=-1$$

$$\textcircled{II} \int \frac{-x}{(x^2+1)} dx = -\frac{1}{2} \int \frac{x}{(x^2+1)} dx = -\frac{1}{2} \int \frac{\cancel{x}}{(u+1)} \frac{du}{2x} = -\frac{1}{2} \int \frac{1}{(u+1)} du = -\frac{1}{2} \ln|x^2+1|$$

$$u = x^2 \quad \frac{du}{dx} = 2x \quad dx = \frac{du}{2x}$$

$$\textcircled{15} \int \sec x dx = \int \frac{\cos(x)}{\cos^2(x)} dx = \int \frac{\cos(x)}{1-\sin^2(x)} dx = \frac{1}{2} \cdot \left(\ln \left| \frac{1+\sin x}{1-\sin x} \right| \right) + C //$$

$$\textcircled{I} \int \frac{\cos(x)}{1-\sin^2(x)} dx = \int \frac{\cos(x)}{1-u^2} dx = \int \frac{\cos(x)}{1-u^2} \cdot \frac{du}{\cos x} = \int \frac{1}{1-u^2} du = \frac{1}{2} \ln|1-u| + \frac{1}{2} \ln|1+u|$$

$$u = \sin(x)$$

$$\frac{du}{dx} = \cos(x) \rightarrow dx = \frac{du}{\cos(x)}$$

$$\textcircled{II} \int \frac{1}{1-u^2} du = \int \frac{1}{(1-u) \cdot (1+u)} du = \int \frac{A}{(1-u)} du + \int \frac{B}{(1+u)} du = \int \frac{1/2}{(1-u)} du + \int \frac{1/2}{(1+u)} du = \frac{1}{2} \ln|1+\sin x| - \frac{1}{2} \ln|1-\sin x| = \frac{1}{2} \cdot \left(\ln \left| \frac{1+\sin x}{1-\sin x} \right| \right) + C$$

$$\frac{1}{(1-u) \cdot (1+u)} = \frac{A}{(1-u)} + \frac{B}{(1+u)} = \frac{A(1+u) + B(1-u)}{(1-u) \cdot (1+u)} = \frac{A + Au + B - Bu}{(1-u) \cdot (1+u)} = \frac{u \cdot (A-B) + A+B}{(1-u) \cdot (1+u)}$$

$$\begin{aligned} A-B &= 0 & A &= \frac{1}{2} \\ u \cdot (A-B) + A+B &= 1 \Rightarrow & A+B &= 1 & B &= \frac{1}{2} \end{aligned}$$

$$\textcircled{III} \int \frac{1/2}{(1-u)} du = \frac{1}{2} \int \frac{1}{1-u} du = \frac{1}{2} \int -\frac{1}{x} dx = -\frac{1}{2} \int \frac{1}{x} dx = -\frac{1}{2} \ln|x| = -\frac{1}{2} \ln|1-u| = -\frac{1}{2} \ln|1-\sin x|$$

$$x = 1-u \quad \frac{du}{dx} = -1 \quad du = -dx$$

$$\textcircled{IV} \int \frac{1/2}{(1+u)} du = \frac{1}{2} \int \frac{1}{1+u} du = \frac{1}{2} \int \frac{1}{q} dq = \frac{1}{2} \ln|q| = \frac{1}{2} \ln|1+u| = \frac{1}{2} \ln|1+\sin x|$$

$$q = 1+u \quad \frac{dq}{du} = 1 \quad dq = du$$

COMPLEMENTO DE QUADRADOS:

$$17) a) \int \frac{x^3 - 2x}{x^2 - 1} dx = \int x - \frac{x}{x^2 - 1} dx = \int x dx - \int \frac{x}{x^2 - 1} dx = \frac{x^2}{2} - \frac{\ln|x^2 - 1|}{2} + C$$

$$\begin{array}{r} x^3 + 0x^2 - 2x \\ -x^2 + 0x + 1 \\ \hline x^3 - x^2 - 2x + 1 \end{array} \quad \begin{array}{r} x^2 + 0x - 1 \\ -x^2 + 0x + 1 \\ \hline 0 \end{array} \rightarrow x - \frac{x}{x^2 - 1}$$

$$\textcircled{I} \int \frac{x}{x^2 - 1} dx = \int \frac{1}{u - 1} \frac{du}{2} = \frac{\ln|u - 1|}{2} = \frac{\ln|x^2 - 1|}{2}$$

$$u = x^2 \quad \frac{du}{dx} = 2x \quad \frac{du}{2} = x dx$$

$$b) \int \frac{x}{x^2 - 4x + 8} dx = \int \frac{x}{(x-2)^2 + 4} dx = \int \frac{u+2}{u^2 + 4} du = \frac{1}{2} \ln|(x-2)^2 + 4| + \arctan\left|\frac{x-2}{2}\right| + C$$

$$x^2 - 4x = -8 \quad 2 \cdot (-2)$$

$$u = x - 2$$

$$x = u + 2$$

$$x^2 - 4x + 4 = -8 + 4$$

$$(x-2)^2 + 4 = 0$$

$$\int \frac{1}{u^2 + a^2} du = \frac{1}{a} \arctan\left|\frac{u}{a}\right| + C$$

$$\textcircled{I} \int \frac{u+2}{u^2 + 4} du = \int \frac{u}{u^2 + 4} du + \int \frac{2}{u^2 + 4} du = \frac{1}{2} \ln|(x-2)^2 + 4| + \arctan\left|\frac{x-2}{2}\right| + C$$

$$\textcircled{II} \int \frac{u}{u^2 + 4} du = \int \frac{u}{u^2 + 4} \frac{du}{2u} = \frac{1}{2} \int \frac{1}{u^2 + 4} du = \frac{1}{2} \ln|u^2 + 4| = \frac{1}{2} \ln|(x-2)^2 + 4| + C$$

$$u = x^2 \quad \frac{du}{dx} = 2x \quad \frac{du}{2u} = \frac{dx}{2x}$$

$$\textcircled{III} \int \frac{2}{u^2 + 4} du = 2 \cdot \frac{1}{2} \arctan\left|\frac{u}{2}\right| = \arctan\left|\frac{x-2}{2}\right| + C$$

$$e) \int \frac{x^2 - 4}{x - 1} dx = \int (x+1) + \frac{-3}{x-1} dx = \int (x+1) dx + \int \frac{-3}{x-1} dx = \frac{x^2 + 2x}{2} - 3 \ln|x-1| + C$$

$$\begin{array}{r} x^2 + 0x - 4 \\ -x^2 + 1x \\ \hline 1x - 4 \\ -1x + 1 \\ \hline -3 \end{array}$$

$$\textcircled{I} \int (x+1) dx = \int x dx + \int 1 dx = \frac{x^2}{2} + x = \frac{x^2 + 2x}{2} + C$$

$$\textcircled{II} \int \frac{-3}{x-1} dx = -3 \int \frac{1}{x-1} dx = -3 \ln|x-1|$$

$$d) \int \frac{2x^2 - 1}{(4x-1)(x^2+1)} dx = * \text{CONTINUAR}$$

$$g) \int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} dx = \int \frac{(x-2)(x+1)}{(3x-1)(x^2+1)} dx = \int \frac{-7}{5} \cdot \frac{1}{(3x-1)} dx + \int \frac{\frac{4}{5}x + \frac{3}{5}}{x^2+1} dx = \frac{-7}{5} \int \frac{1}{3x-1} dx + \int \frac{\frac{4}{5}x}{x^2+1} dx + \int \frac{\frac{3}{5}}{x^2+1} dx = *$$

$$\begin{array}{r} 3x^3 - x^2 + 3x - 1 \\ x^2(3x-1) + (3x-1) \\ \hline (3x-1)(x^2+1) \end{array}$$

$$= * \int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} = \frac{-7}{15} \ln|x-1| + \frac{2}{5} \ln|x^2+1| + \frac{3}{5} \arctan(x)$$

$$\frac{(x-2)(x+1)}{(3x-1)(x^2+1)} = \frac{A}{(3x-1)} + \frac{Bx+C}{(x^2+1)} = \frac{A(x^2+1) + (Bx+C)(3x-1)}{(3x-1)(x^2+1)} = \frac{Ax^2 + A + 3Bx^2 - Bx + 3Cx - C}{(3x-1)(x^2+1)} = \frac{x^2(A+3B) + x(-B+3C) + A-C}{(3x-1)(x^2+1)} = \frac{x^2 + x - 2}{(3x-1)(x^2+1)}$$

$$\begin{cases} A+3B=1 \\ 3C-B=1 \\ A-C=-2 \end{cases} \rightarrow \begin{cases} C = \frac{6}{10} = \frac{3}{5} \\ B = \frac{8}{10} = \frac{4}{5} \\ A = -\frac{7}{5} \end{cases} \rightarrow \frac{(x-2)(x+1)}{(3x-1)(x^2+1)} = \frac{-\frac{7}{5}}{3x-1} + \frac{\frac{4}{5}x + \frac{3}{5}}{(x^2+1)} = \frac{-7}{5} \cdot \frac{1}{(3x-1)} + \frac{\frac{4}{5}x + \frac{3}{5}}{(x^2+1)} \quad * \text{COLOCAR DE VOLTA NA INTEGRAL}$$

$$\textcircled{I} \int \frac{1}{3x-1} dx = -\frac{1}{3} \int \frac{1}{u-1} \cdot \frac{du}{3} = -\frac{1}{9} \ln|x-1|$$

$$u=3x \quad \frac{du}{dx}=3$$

$$\textcircled{II} \int \frac{4x}{x^2+1} = \int \frac{4}{5} \cdot \frac{x}{x^2+1} dx = \frac{4}{5} \int \frac{x}{x^2+1} dx = \frac{4}{5} \int \frac{\cancel{x}}{u} \cdot \frac{du}{2\cancel{x}} = \frac{2}{5} \ln|x^2+1|$$

$$u=x^2+1 \quad \frac{du}{dx}=2x \quad dx=\frac{du}{2}$$

$$\textcircled{III} \int \frac{3}{x^2+1} = \frac{3}{5} \int \frac{1}{x^2+1} dx = \frac{3}{5} \arctg(x)$$

$$b) \int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2} dx = \int 3x^2 dx + \int 1 dx + \int \frac{1}{x^2+x-2} dx = x^3+x+\frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + C //$$

$$\begin{array}{r} 3x^4+3x^3-5x^2+x-1 \\ -3x^4-3x^3+6x^2 \\ \hline x^2+x-1 \\ -x^2-x+2 \\ \hline 1 \end{array} \quad \begin{array}{l} x^2+x-2 \\ 3x^2+1 \end{array} = 3x^2+1 + \frac{1}{x^2+x-2}$$

$$\textcircled{I} \int \frac{1}{x^2+x-2} dx = \int \frac{1}{(x+2)(x-1)} dx = \int \frac{\frac{-1}{3}}{(x+2)} dx + \int \frac{\frac{2}{3}}{(x-1)} dx = -\frac{1}{3} \int \frac{1}{x+2} dx + \frac{2}{3} \int \frac{1}{x-1} = \frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + C$$

$$\frac{1}{(x+2)(x-1)} = \frac{A}{(x+2)} + \frac{B}{(x-1)} = \frac{A(x-1)+B(x+2)}{(x+2)(x-1)} = \frac{Ax-A+Bx+2B}{(x+2)(x-1)} = \frac{x(A+B)+2B-A}{(x+2)(x-1)}$$

Stesso netto pero o principal.

$$x(A+B)+2B-A=1+0x^2$$

$$\begin{cases} A+B=0 & A=-\frac{1}{3} \\ 2B-A=1 & B=\frac{1}{3} \end{cases} \longrightarrow \frac{A}{(x+2)} + \frac{B}{(x-1)} = \frac{-\frac{1}{3}}{(x+2)} + \frac{\frac{2}{3}}{(x-1)} \quad * \text{Aggero vai de netto pero o integral.}$$

$$h) \int \frac{x^2}{x^2-3x+2} dx = \int \frac{x^2}{(x-2)(x-1)} dx = \int 1 dx + \int \frac{3x-2}{x^2-3x+2} dx = x + \int \frac{4}{x-2} dx - \int \frac{1}{x-1} = x+4 \ln|x-2| - \ln|x-1| + C$$

$$\begin{array}{r} x^2+0x+0 \\ -x^2+3x-2 \\ \hline 3x-2 \end{array} \quad \begin{array}{l} x^2-3x+2 \\ 1 \end{array} \longrightarrow 1 + \frac{3x-2}{x^2-3x+2}$$

$$\textcircled{I} \frac{3x-2}{(x-2)(x-1)} = \frac{A}{(x-2)} + \frac{B}{(x-1)} = Ax-A+Bx-2B = x(A+B)-A-2B$$

$$3x-2=x(A+B)-A-2B$$

$$\begin{cases} A+B=3 & A=4 \\ -A-2B=-2 & B=-1 \end{cases} \longrightarrow \frac{4}{x-2} - \frac{1}{x-1}$$

$$i) \int \frac{1}{3x^2 - 2x + 5} dx = \int \frac{1}{3(x^2 - \frac{2x}{3} + \frac{5}{3})} dx = \frac{1}{3} \int \frac{1}{(x - \frac{1}{3})^2 + \frac{14}{9}} dx = \frac{1}{\sqrt{14}} \cdot \arctan \left| \frac{3x-1}{\sqrt{14}} \right| + C$$

$$x^2 - \frac{2x}{3} + \frac{1}{9} = -\frac{5}{3} + \frac{1}{9}$$

$$(x - \frac{1}{3})^2 + \frac{14}{9} = -\frac{5}{3} + \frac{1}{9}$$

$$(x - \frac{1}{3})^2 + \frac{14}{9} = x^2 - \frac{2x}{3} + \frac{1}{9}$$

$$\text{* sabemos que } \int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan \left| \frac{u}{a} \right| + C$$

$$\textcircled{I} \frac{1}{3} \int \frac{1}{(x - \frac{1}{3})^2 + \frac{14}{9}} dx = \frac{1}{3} \int \frac{1}{(u)^2 + \frac{14}{9}} du = \frac{1}{3} \int \frac{1}{u^2 + (\sqrt{\frac{14}{9}})^2} du = \frac{1}{3} \cdot \frac{3}{\sqrt{14}} \arctan \left| \frac{u}{\sqrt{\frac{14}{9}}} \right| = \frac{1}{\sqrt{14}} \arctan \left| \frac{3x-1}{\sqrt{14}} \right| + C$$

$$u = x - \frac{1}{3} \quad du = dx$$

SUBSTITUIÇÃO TRIGONOMETRICA:

$$d) \int_1^{\sqrt{2}} \frac{1}{x^2 \sqrt{4-x^2}} dx = \int \frac{2 \cos \theta d\theta}{x^2 \sqrt{2^2 - (2 \sin \theta)^2}} = \int \frac{2 \cos \theta d\theta}{x^2 \sqrt{2^2 (1 - \sin^2 \theta)}} = \int \frac{2 \cos \theta d\theta}{x^2 \sqrt{2^2 \cos^2 \theta}} = \int \frac{2 \cos \theta d\theta}{x^2 \cdot 2 \cos \theta} = \int \frac{1}{4 \sin^2 \theta} d\theta = -\frac{1}{4} \cot \theta = *$$

$$\sqrt{2^2 - x^2} \quad \begin{array}{c} 2 \\ \theta \\ 0 \end{array} \quad x \quad \boxed{0 = \sqrt{2^2 - x^2}}$$

$$\sin \theta = \frac{x}{2} \quad \therefore x = 2 \sin \theta$$

$$* = -\frac{1}{4} \frac{\cos \theta}{\sin \theta} = -\frac{1}{4} \frac{\sqrt{2^2 - x^2}}{\frac{x}{2}} = \left[-\frac{1}{4} \frac{\sqrt{2^2 - x^2}}{x} \right]_1^{\sqrt{2}} = \left| -\frac{1}{4} \right| - \left| -\frac{\sqrt{3}}{4} \right| = \frac{\sqrt{3}-1}{4} //$$

$$\frac{dx}{d\theta} = 2 \cos \theta \quad dx = 2 \cos \theta d\theta$$

$$\sin \theta = \frac{x}{2} \quad \cos \theta = \frac{\sqrt{2^2 - x^2}}{2}$$

$$\text{Para } x = \sqrt{2} = \frac{\sqrt{2}}{2} = 1 \cdot \left(\frac{1}{2} \right) = -\frac{1}{4}$$

$$\text{Para } x = 1 = -\frac{1}{4} \cdot \frac{\sqrt{3}}{1} = -\frac{\sqrt{3}}{4}$$

$$b) \int \frac{\sqrt{x^2 - 25}}{x} dx = \int \frac{5 \tan \theta}{5 \sec \theta} dx = \int \frac{\tan \theta}{\sec \theta} 5 \sec \theta \tan \theta d\theta = 5 \int \tan^2 \theta d\theta = 5 \int (\sec^2 \theta - 1) d\theta = 5 [\tan \theta - \theta] = \sqrt{x^2 - 25} - 5 \arccos \left(\frac{5}{x} \right)$$

$$\textcircled{I} \sqrt{x^2 - 5^2} \quad \begin{array}{c} x \\ \theta \\ 5 \end{array} \quad y = \sqrt{x^2 - 5^2} \quad \tan \theta = \frac{\sqrt{x^2 - 5^2}}{5}$$

$$\cos \theta = \frac{5}{x} \quad x = 5 \sec \theta \quad \therefore \sqrt{x^2 - 5^2} = \sqrt{(5 \sec \theta)^2 - 5^2} = \sqrt{5^2 \sec^2 \theta - 5^2} = \sqrt{5^2 (\sec^2 \theta - 1)} = 5 \tan \theta$$

$$\boxed{\theta = \arccos \left(\frac{5}{x} \right)} \quad \frac{dx}{d\theta} = 5 \sec \theta \tan \theta \quad \therefore dx = 5 \sec \theta \tan \theta d\theta$$

$$\int \frac{\sqrt{x^2 - 25}}{x} dx = \sqrt{x^2 - 25} - 5 \arccos \left(\frac{5}{x} \right) + C$$

$$c) \int \sqrt{4-x^2} dx = \int \sqrt{2^2 - x^2} \cdot 2 \cos \theta d\theta = \int \sqrt{2^2 (1 - \sin^2 \theta)} \cdot 2 \cos \theta d\theta = \int 2 \cos \theta \cdot 2 \cos \theta d\theta = 4 \int \cos^2 \theta d\theta = 2 \int (1 + \cos 2\theta) d\theta = 2 \left[\theta + \frac{\sin 2\theta}{2} \right] = *$$

$$\sqrt{4-x^2} \quad \begin{array}{c} 2 \\ \theta \\ 0 \end{array} \quad x \quad y = \sqrt{4-x^2}$$

$$* 2 \left[\theta + \frac{\sin 2\theta}{2} \right] = 2 \arcsin \left(\frac{x}{2} \right) + x \sqrt{4-x^2}$$

$$\sin \theta = \frac{x}{2} \quad \therefore x = 2 \sin \theta$$

$$\theta = \arcsin \left(\frac{x}{2} \right)$$

$$\textcircled{I} \int \cos 2\theta d\theta = \int \cos u \frac{du}{2} = \frac{\sin 2\theta}{2}$$

$$u = 2\theta \quad \frac{du}{d\theta} = 2 \quad d\theta = \frac{du}{2}$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta = 2 \cdot \frac{x}{2} \cdot \frac{\sqrt{4-x^2}}{2}$$

$$\frac{dx}{d\theta} = 2 \cos \theta \quad \therefore dx = 2 \cos \theta d\theta$$